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Mathias Louboutin

Summary

I have recently graduated with a Ph.D from Georgia Institute of Technology. My doctoral work focused on domain-specific language for large scale finite-difference solvers in seismic exploration. My research includes large scale PDE constrained optimization, finite-difference solver, non-linear and non-convex optimization with recent extensions to medical imaging. The second field of my research is to apply recent machine learning techniques to seismic exploration for applications such as missing data interpolation (infilling), numerical dispersion removal and seismic data processing.

Experience

July 2020–* **PostDoctoral Fellow with Pr. Felix J. Herrmann. School of Earth Science, CSE**, *Atlanta*, USA, Seismic Laboratory for imaging and modelling (SLIM).

High performance computing on Azure for exascale PDE constraints inverse problems. Machine learning augmented wave-equation based inversion. Wave-equation based medical imaging. High performance finite-difference modeling.

Selected publications

Devito (v3.1.0): an embedded domain-specific language for finite differences and geophysical exploration, 2018, *Geoscientific Model Development*, `https://doi.org/10.5194/gmd-12-1165-2019`.

Scaling through abstractions - high-performance vectorial wave simulations for seismic inversion with Devito, 2020, *Technical report*, `https://slim.gatech.edu/Publications/Public/TechReport/2020/louboutin2020SCsta/louboutin2020SCsta.html`.

An Event-Driven Approach to Serverless Seismic Imaging in the Cloud, 2019, *Submitted to IEEE Transactions on Parallel and Distributed Systems on August 20, 2019*, `https://slim.gatech.edu/Publications/Public/Submitted/2019/witte2019TPDedas/witte2019TPDedas.html`.

Neural network augmented wave-equation simulation, 2019, *Technical report*, `https://arxiv.org/abs/1910.00925`.

The importance of transfer learning in seismic modeling and imaging, 2019, *GEOPHYSICS*, `http://dx.doi.org/10.1190/geo2019-0056.1`.

Architecture and performance of Devito, a system for automated stencil computation, 2018, *Submitted to SIAM Journal on Scientific Computing on July 9, 2018.*, `https://arxiv.org/abs/1807.03032`.

Extending the search space of time-domain adjoint-state FWI with randomized implicit time shifts, 2017, *EAGE Paris*, `http://dx.doi.org/10.3997/2214-4609.201700831`.

Performance prediction of finite-difference solvers for different computer architectures, 2016, *Computer and Geoscience*, `https://doi.org/10.1016/j.cageo.2017.04.014`.

Education

2018–May 2020 **Ph.D student at Georgia Institute of Technology under the supervision of Pr. Felix J. Herrmann. School of Computer Science, CSE**, *Atlanta*, USA, Seismic Laboratory for imaging and modelling (SLIM).

Numerical and computational methods for large scale simulation based inverse problems and machine learning.

2013–2018 **Ph.D student at University of British Columbia under the supervision of Pr. Felix J. Herrmann**, *Vancouver*, Canada, Seismic Laboratory for imaging and modelling (SLIM).

Full-waveform inversion, Exascale computational geophysics.

2016 Feb-Aug **Visiting Ph.D student at Imperial college under the supervision of Pr. Gerard Gorman**, *London*, UK.

Automatic code generation for geophysical exploration applications with finite differences.

2011–2013 **Master of Science in Mathematics specializing in modeling**, *Universite de Rennes 1*, Rennes, France.
Well recognized French mathematics Master degree. Equivalent to an American MSc in applied mathematics.

2013 Valedictorian

Required coursework: Calculus, Numerical Methods, PDE Resolution, Optimization, C/C++ Computing, Mathematics Modeling and Simulation, Finite Element Method

Elective coursework: Fluid Mechanics, Continuum Mechanics and Thermomechanics, Bio-mechanics, Geophysics Modeling

2008–2011 **ENSICA, Graduate Institute of Aeronautical Engineering**, *ISAE*, Toulouse, France.
Leading French Aeronautical Engineering School.Two years on three validated, left at the end of the second year.

Required coursework: Mathematics, Mechanics, Continuum Mechanics, Structures Mechanics, Signal Processing, Thermodynamics, Fluid Mechanics, Java Computing

Elective coursework : Estimation Methods, Earth Observation Satellites, Microwaves Processing

Second year team project: Modeling of the dynamics of sand dunes. Theoretical and experimental study in collaboration with a Ph.D student of the Fluids Mechanics Institute of Toulouse(IMFT). Granular and Fluid Mechanics

2006–2008 **Advanced undergraduate coursework in Mathematics and Physics**, *Lycee Chateaubriand*, Rennes, France.

Second year team project: Studied structure and rupture conditions of avalanches. Conducted experimental study on a small-scale model

Intenships

Summer 2013 **Research internship**, *ONERA*, Toulouse, France.
Scattering patterns of atmospheric dust clouds analysis with the Discrete Dipole Approximation (DDA) method.

Summer 2012 **Research internship**, *INRIA*, Grenoble, France.
Intern in NANO-D department at INRIA-Grenoble. Working on L2-SVM: quadratic optimization for large scale problems (millions of constraints, 10,000 unknowns). Dual and primal resolution. Runtime and accuracy improvement on a specific problem.

Summer 2011 **Internship**, *Aeroconseil*, Toulouse, France.
Developed an interface for aerodynamics calculus in JAVA. Reading and implementation of Excel and Scilab scripts through the interface

2006-2013	Managed summer camp for 6-13 years old children. 8 Summer camp with the organism TELLIGO
2006-2013	High-school tutor in Mathematics and Physics

Languages

French	Native speaker
English	advanced level, Doctoral degree at an American university

Computer skills

OS	Linux, Unix, Windows
Programming	Python, C, Julia, Matlab
Cloud computing experience	Azure, AWS, GCP
Miscellaneous	Office, Latex, Markdown, Shell script